



Evaluation of an extruded diet for adult dogs containing larvae meal from the Black soldier fly (*Hermetia illucens*)

Susan Kröger ^{*}, Carolin Heide, Jürgen Zentek

Institute of Animal Nutrition, Department of Veterinary Medicine, Freie Universität Berlin, Königin-Luise- Str. 49, 14195, Berlin, Germany

ARTICLE INFO

Keywords:

Apparent total tract digestibility
Dog
Immune system
Insect protein

ABSTRACT

Insects have experienced an increasing interest as a protein source in recent years. However, their suitability as a protein source for dogs was not yet widely investigated. Therefore, the aim of the study was to compare two extruded diets for adult dogs containing either larvae meal from black soldier fly (*Hermetia illucens*; HI) or lamb meal as the predominant protein source with regard to apparent fecal nutrient digestibility and the influence on fecal and immunological parameters. Twelve Beagles were used in this cross-over study and divided into two groups. The test diet with 200 g/kg HI larvae meal (HI) was compared with a control diet containing lamb meal (CON). Dogs were fed each diet for a period of five weeks. Blood samples for a complete blood count, lymphocyte phenotyping and lymphocyte proliferation tests were taken at the end of each feeding period. The latter was either mitogen- or feed antigen-induced with different protein concentrations of HI larvae meal (30–500 µg protein/mL). Fecal samples were collected for five days during each feeding period. The coefficient of the total apparent fecal digestibility (CTTAD) of macronutrients, fecal metabolites, the output and the dry matter content of the feces were analyzed. D-Glucosamine as indicator for chitin was analyzed by high-performance anion-exchange chromatography with pulsed amperometric detection in the diet and the feces. The chitinase activity was determined in the feces. The fecal output was reduced in dogs fed the HI diet ($P = 0.038$). The CTTAD of dry matter was higher ($P = 0.002$) when dogs were fed HI compared to CON (0.832 vs. 0.818) and the CTTAD of crude protein was higher ($P = 0.002$) when dogs received CON compared to HI (0.792 vs. 0.773). Concentrations of acetate ($P = 0.011$) and ammonium ($P = 0.001$) were higher in CON. The fecal score differed ($P = 0.041$) between HI (2.25) and CON (2.04), but both feeding groups had a formed and dry fecal consistency. The chitin concentration was numerically higher in the feces of dogs fed HI (92.2 g/kg DM) compared to CON (2.9 g/kg DM), the fecal chitinase activity did not differ between both diets. Complete blood counts, lymphocyte phenotyping and proliferation were not affected. In conclusion, the HI larvae meal in a dry dog diet was tolerated without adverse signs and did not affect immunological measurements compared to a commercial dry diet with lamb meal, indicating that larvae meal from HI can be considered an “alternative” protein source for dog nutrition.

Abbreviations: BrdU, 5-bromo-2'-deoxyuridine; CON, control diet; CTTAD, coefficient of the total tract apparent digestibility; HI, *Hermetia illucens*; PBS, Phosphate-buffered saline; PHA, Phytohemagglutinin; PWM, Pokeweed Mitogen; SCFA, short-chain fatty acids; TDF, total dietary fiber.

^{*} Corresponding author.

E-mail addresses: susan.kroeger@fu-berlin.de (S. Kröger), heide.carolin@gmx.de (C. Heide), Juergen.Zentek@fu-berlin.de (J. Zentek).

<https://doi.org/10.1016/j.anifeedsci.2020.114699>

Received 11 May 2019; Received in revised form 19 March 2020; Accepted 19 September 2020

Available online 8 October 2020

0377-8401/© 2020 Elsevier B.V. All rights reserved.

1. Introduction

In search of alternative protein sources, insect proteins have received increasing interest in the last decade. The black soldier fly (*Hermetia illucens*; HI) is one of the most relevant species cultured. The reported protein concentrations of larvae meals vary between 362 g/kg (Barroso et al., 2014) and 561 g/kg (Bosch et al., 2014) and can be compared to other protein sources usually used in dog food such as meats or fish meal. The native fat content of insect larvae can be high (Rumpold and Schlüter, 2013), with values in partially defatted HI larvae meals being between 128 g/kg (Bosch et al., 2014), 180 g/kg (Barroso et al., 2014) and 348 g/kg (Newton et al., 1977). In addition to the nutritional value, insect protein has some ecological advantages. Compared with conventional meat production the production of insects produces less greenhouse gases and ammonia (Oonincx et al., 2010). There are also some economic aspects, as insects can be kept on a broad variety of organic materials, although bioconversion can be variable depending on the substrate quality. Compared to the production of protein from soy or fat from rapeseed the production of protein and fat from HI showed beneficial environmental impacts, but it still has some disadvantages as it needs a large amount of energy for the production (Salomone et al., 2017).

The suitability of HI larvae meal for dogs and intestinal physiological effects has not been widely studied in *in vivo* trials. Our hypothesis was, that, based on its nutrient composition and previous reports (Bosch et al., 2014), HI larvae meal would have a comparable apparent total tract digestibility in the intestinal tract as classical animal protein sources, that dogs can tolerate insect-based diets without negative impact on the digestive function and on immunological traits. The latter is based on the fact that to the best of our knowledge there are no reports of intolerances or allergy in dogs, even if this does not exclude the possibility of influencing the immune system, for example also through chitin, which consists of acetyl-glucosaminoglycans.

2. Material and methods

The study was approved by the local state office of occupational health and technical safety 'Landesamt für Gesundheit und Soziales Berlin' (LaGeSo Reg. Nr. 0195/15).

2.1. Animals, diets and sampling

A total of twelve Beagles (six males and six females) with an age of 28 ± 10 months were involved in a cross-over study. The dogs were divided into two groups blocked for age and sex and housed in pairs. During the sampling period the dogs were housed individually.

Two extruded complete dry foods were used (provided by Vet-Concept GmbH, Föhren Germany): the test diet with HI larvae meal (HI) as the primary protein source was compared with a reference diet containing lamb meal (CON) as main protein source (Table 1). Both diets were formulated to be iso-nitrogenic, however, other nutrients were not completely comparable (Table 2).

The concentration of total dietary fiber (TDF), insoluble dietary fiber and soluble dietary fiber in both diets were analyzed according to the assay procedure K-TDFR-100A/K-TDFR-200A 04/17 of the manufacturer Megazyme (Ireland).

The diets did not only differ in the protein carrier but also in some other components (Table 2). The diet with lamb meal has been on the market for many years as a standard diet for dogs and is known to be very well tolerated by sensitive dogs. Therefore, this diet was used as a reference against the diet with HI larvae meal.

The individual maintenance energy requirement was calculated for each dog according to the recommendations of the National Research Council (National Research Council (NRC, 2006) for adult dogs and the diets were offered once a day to cover estimated maintenance requirements. When necessary, the daily food allowance was adjusted to achieve constancy of body weight. All dogs had free access to tap water.

The dogs were fed each diet for a period of five weeks. At the last five days of each feeding period dogs were housed individually for total fecal collection, determination of fecal score and feces amount. Daily aliquots were taken, pooled and homogenized for each dog

Table 1

Analyzed chemical composition of the predominant protein sources lamb meal and *Hermetia illucens* larvae meal ; data provided by Vet-Concept GmbH, Föhren Germany.

Nutrients, g/kg, as fed	Diet	
	Lamb meal	HI ^a
Moisture	48.0	41.0
Crude protein	582	618
Acid hydrolyzed ether extract	123	65.0
Crude ash	226	76.0
Crude fiber	12.0	103
Calcium	76.3	8.60
Phosphorus	39.0	13.6
Potassium	5.30	18.4
Magnesium	2.00	5.10
Sodium	7.70	1.20

^a HI = *Hermetia illucens* larvae meal.

Table 2Analyzed chemical composition and ingredients of the control diet with lamb meal and the diet with *Hermetia illucens*.

Nutrients, g/kg dry matter	Diet	
	CON ^a	HI ^a
Crude protein	269	260
Acid hydrolyzed ether extract	90.8	74.9
Crude ash	84.5	54.8
Crude fiber	14.3	23.6
Total dietary fiber	60.0	51.0
Insoluble dietary fiber	52.1	41.6
Soluble dietary fiber	7.90	9.40
Calcium	19.7	8.40
Phosphorus	12.8	10.3
Potassium	3.4	6.1
Magnesium	1.0	1.5
Sodium	3.7	2.4
Ingredients	Rice meal, lamb meal, barley, rice protein, sugar beet pulp, poultry fat, tallow, sunflower oil, flax seed, rape seed oil, salt, yeast (extracted), vitamin/mineral premix	Rice meal, HI larvae meal, barley, potato protein, mono-calcium-phosphate, poultry fat, tallow, sunflower oil, salt, limestone, rape seed oil, yeast (extracted), vitamin/mineral premix

^a CON = control diet with lamb meal; HI = diet with *Hermetia illucens*.

to determine the apparent total tract nutrient digestibility, fecal dry matter, concentration of chitin as well as chitinase activity. In addition, fresh feces samples were taken for analysis of microbial metabolites. Blood samples were taken at the last day of each feeding period for lymphocyte phenotyping and lymphocyte proliferation tests and the performance of complete and differential blood counts.

2.2. Fecal consistency

The fecal consistency was assessed daily during the five-day sampling period. For the determination a subjective 5-point scoring system was used (Middelbos et al., 2007): 1 = hard, dry pellets; small hard mass; 2 = hard-formed, dry stool, remains firm and soft; 3 = soft, formed, and moist stool, retains shape; 4 = soft, unformed stool; assumes shape of container; 5 = watery, liquid that can be poured.

2.3. Total tract apparent nutrient digestibility

The total tract apparent digestibility of nutrients was determined using 0.2 % titanium dioxide (Sigma Aldrich, St. Louis, USA) as an indigestible marker in both diets. Proximate nutrients (crude protein, crude ash, ether extract) and dry matter were analyzed in the diets and the feces according to standard procedures (Naumann and Bassler, 2004). Organic matter was calculated after dry ashing at 550 °C during 6 h and N-free extracts were calculated according to the official methods (Naumann and Bassler, 2004). Titanium dioxide was analyzed photometrically according to Short et al. (1996). The following formula was used to calculate the coefficient of the total tract apparent digestibility (CTTAD): $100 - [(\% \text{ indicator in feed} / \% \text{ indicator in feces}) * (\% \text{ nutrient in feces} / \% \text{ nutrient in feed}) * 100]$

2.4. D-glucosamine concentration as marker for chitin in diet and feces

The chitin concentration in the CON and HI diet as well as in the feces was estimated by the determination of D-glucosamine after acid and alkaline hydrolysis of samples. D-glucosamine is not specific for chitin and is also found in other animal tissues. Therefore, the test is ultimately to be considered a useful marker, but not specific for the chitin content.

Diets and fecal aliquots (700 mg) were ground (0.5 mm) and weighed into filter bags (F57, ANKOM Technology Macedon, New York, USA). The samples were defatted with petroleum ether and treated successively with sulfuric acid (0.128 mol/L) and sodium hydroxide (0.313 mol/L) at 100 °C in an Ankom 2000 Fiber Analyzer (ANKOM Technology Macedon, New York, USA). Residues were dehydrated with acetone and dried for 45 min at 104 °C. After incubation with 4 mL sulfuric acid (12 mol/L) for 60 min at 40 °C, 30 mL ultrapure water was added and incubated at 100 °C for 2 h. Samples were cooled off, the pH was adjusted to 8–9 with sodium hydroxide (7.5 mol/L) and the suspension was filled up to 100 mL with ultrapure water. After filtration, 5 µL of the samples were analyzed by high-performance anion-exchange chromatography with pulsed amperometric detection on a CarboPac PA1 column with alkaline gradient elution (Supplementary Table 1). D-glucosamine was quantified by using a calibration curve with concentrations from 10 to 80 µg/mL. For the preparation of the calibration curve, pure chitin derived from shrimp shells (Sigma Aldrich, St. Louis, USA) was used. The same chitin was added to the diet samples as well to the larvae meal alone in a defined amount. Recovery after passing the whole preparation procedure was between 913 and 947 g/kg.

2.5. Chitinase activity in feces

The chitinase activity was determined according to Shen et al. (2010). Briefly, by adding diluted fecal samples to colloid chitin, chitin-azure was converted by fecal chitinase and the released azure dye was determined by photometric detection at 560 nm with Ultrospec 3300 pro and Swift II Quantification software (Amersham Biosciences Corp., Piscataway, USA). In the results one U stands for the release of 1 mg D-Glucosamine of chitin within 1 h.

2.6. Determination of microbial metabolites in fecal samples

Samples for the analysis of microbial metabolites were stored immediately until analysis at -80°C . Short chain fatty acids (SCFA), D- and L-lactate and ammonium were analyzed as described before (Kröger et al., 2017). Briefly, SCFA were analyzed by gas chromatography (Agilent Technologies 6890 N, auto sampler G2614A and injection tower G2613A, Network GC Systems, Böblingen, Germany) with flame ionisation detector. Caproic acid was used as an internal standard (0.5 mmol/L).

The analysis of D- and L-lactate was performed by high performance liquid chromatography (Agilent 1100, Agilent Technologies, Böblingen, Germany) with a pre-column (Phenomenex C18 4.0 L mm x 2.0 ID mm, Aschaffenburg, Germany) and an analytical column (Phenomenex Chirex 3126 (D)-penicillamine 150×4.6 mm, Aschaffenburg, Germany).

Ammonium was analyzed colorimetrically by using the Berthelot-Reaction in microtitration plates. Extinction was measured at 620 nm in a Tecan SunriseTM microplate reader (Tecan Austria GmbH, Grödig, Austria).

2.7. Complete blood count

Blood was taken from *V. cephalica* or *V. antebrachii* and collected in EDTA (2 mL) and lithium heparin tubes (10 mL). Samples were sent for further analysis of complete blood count to an external laboratory using laser flow cytometry and peroxidase staining (Laboklin GmbH & Co. KG, Bad Kissingen, Germany).

2.8. Isolation of peripheral blood mononuclear cells using density gradient centrifugation

For phenotyping and the proliferation test of lymphocytes mononuclear cells were isolated from heparinized blood using Ficoll (Biocoll, Biochrom AG, Berlin, Germany). The blood was diluted 1:2 with phosphate-buffered saline (PBS) and 9 mL were transferred in a vial with 3 mL Ficoll and centrifuged for 30 min at room temperature and $500 \times g$. The interphase which contains the mononuclear cells was removed carefully and transferred to 10 mL PBS, followed by two washing steps with PBS. The pellet was removed, re-suspended with 1 mL and incubated at room temperature for 5 min with 10 mL erythrocyte lysis buffer (108 mL ultrapure water and 12 mL red blood cell lysis solution (Miltenyi Biotech GmbH, Bergisch Gladbach, Germany)). Subsequently the solution was centrifuged for 10 min at 4°C and $300 \times g$ and the pellet was re-suspended in 5 mL cell culture medium, which consists of 500 mL VLE RPMI 1640, 50 mL fetal calf serum and 5 mL penicillin/streptomycin (all Biochrom AG, Berlin, Germany).

2.9. Extraction and determination of protein from *Hermetia illucens* larvae meal

The protein extract was used for the feed antigen-induced lymphocyte proliferation test. For the protein extraction 5 g larvae meal were diluted with 20 mL of Sörensen-sodium phosphate buffer (20 mM and pH 7.4: 10 mL of 0.2 mol/L sodium-dihydrogen-phosphate and 380 mL of 0.02 mol/L disodium-hydrogen-phosphate). The solution was stored on ice, homogenized for 4×30 s and centrifuged for 15 min at 4°C and $20,000 \times g$. The supernatant was centrifuged for 10 min and the second supernatant was again centrifuged for 20 min, each at the same conditions as described before. The resulting supernatant was filtered (folded paper filter), centrifuged for another 10 min and filtered ($0.45 \mu\text{m}$) followed by a sterile filtration ($0.2 \mu\text{m}$). The protein extracts were stored at -25°C .

The protein concentration was determined by the Bradford method (Bradford, 1976) using a bovine serum albumin calibration curve (10–150 $\mu\text{g/mL}$). Samples were diluted 1:20 and 1:25, both in dual approaches. From each pre-diluted sample 100 μL were diluted with 1 mL Bradford solution (20 mg Coomassie brilliant blue G-250, 25 mL 96 % ethanol, 50 mL o-phosphoric acid, and 425 mL ultrapure water). The solution was incubated for 3–10 min and photometrically measured at 595 nm (Ultrospec 3300 pro, Amersham Biosciences Corp., Piscataway, USA). Swift II quantification software (Amersham Biosciences Corp., Piscataway, USA) was used to

Table 3

Fecal score, dry matter and daily output amount of the dogs fed the control diet and the diet containing *Hermetia illucens*^a.

Item	Diet		SEM	P-value
	CON ^b	HI ^b		
Fecal score ^c	2.04	2.25	0.055	0.041
Fecal dry matter, %	35.4	34.3	0.706	0.086
Daily feces amount, g/24 h	137	109	9.4	0.038

^a n = 12 adult dogs per treatment ; six male and six female.

^b CON = control diet with lamb meal; HI = diet with *Hermetia illucens*.

^c 1 (hard) – 5 (liquid) (Middelbos et al., 2007).

create calibration curve.

2.10. Lymphocyte phenotyping

The isolated mononuclear cells were adjusted to a concentration of 1×10^7 /mL by adding cell culture medium. Of this solution 100 μ L (1×10^6 cells) were added to 50 μ L pre-diluted specific antibodies: anti-CD4, anti-CD8 β , anti-CD3, anti-CD5 and anti-MHC II (Table 3). After incubation on ice for 25 min non-binding antibodies were washed out by adding 2 mL FACS buffer (1000 mL PBS and 5 g albumin fraction V [Carl Roth GmbH & Co. KG, Karlsruhe, Germany]) and centrifugation for 5 min at 4 °C and 400 x g. The pellet was re-suspended with 50 μ L pre-diluted secondary antibodies (Supplementary table 2) and incubated on ice for 20 min. Non-binding antibodies were washed out and centrifuged again as described before and the remaining pellet was re-suspended for flow cytometric measurement with 200 μ L FACS buffer. For every dog a non-stained sample served as a negative control. Compensation was performed to avoid overlap of the different channels (FITC and PC) to avoid false positive signals.

2.11. Lymphocyte proliferation test

The lymphocyte proliferation test was either mitogen-induced (Pokeweed Mitogen, PWM and Phytohemagglutinin, PHA; both Sigma Aldrich, St. Louis, USA) or feed antigen-induced with different protein concentrations of larvae meal (30, 75, 125, 250, and 500 μ g protein/mL).

The isolated mononuclear cells were adjusted to a concentration of 4×10^6 /mL by adding cell culture medium and 10 μ L were mixed with 100 μ L pre-diluted PWM (2.5 μ g/mL) or PHA (10 μ g/mL). The samples were incubated under CO₂ conditions for 48 h before 20 μ L 5-bromo-2-desoxyuridine (BrdU, Sigma Aldrich, St. Louis, USA) was added. After another CO₂ incubation for 24 h samples were centrifuged for 5 min at 4 °C and 500 x g and the supernatant was removed and frozen. The remaining pellet was re-suspended in 1 mL FACS buffer, followed by a fixation with "Inside Fix" (250 μ L; Miltenyi Biotec GmbH, Bergisch Gladbach, Germany). Mixed carefully at a low frequency and incubated at room temperature for 10 min.

The data of both lymphocyte phenotyping tests are given as stimulation index (SI) = proliferation of cells treated with mitogen / proliferation of cells cultured without mitogen.

2.12. Statistical analysis

Statistical analysis was performed with SPSS (2017). Data were analyzed with the general linear model considering between group effects as well as within subject effects (repeated measures). The level of significance was set at $P < 0.05$, and $0.05 < P < 0.1$ was considered as a trend. For variables means and the SEM were reported.

3. Results

3.1. Fecal consistency, fecal dry matter and daily feces amount

The fecal score was greater for dogs fed the HI diet compared to the CON diet (Table 3) ($P = 0.041$), but feces were formed during both treatments. Fecal dry matter was not different between the HI and the CON diet (Table 3) and the daily fecal output was reduced ($P = 0.038$) when dogs were fed the HI diet (Table 3).

3.2. Total tract apparent nutrient digestibility

The CTTAD was similar between the two test diets. Minor differences were seen for dry matter ($P = 0.002$) and nitrogen-free extract ($P < 0.001$) being higher and crude protein being lower ($P = 0.002$) with the HI diet than with CON (Table 4).

Table 4

Coefficient of the total tract apparent digestibility of dry matter, organic matter, crude protein, ether extract, crude ash, crude fiber, and N-free extracts of dogs fed the control diet (CON) and the diet containing *Hermetia illucens*^a.

Item, coefficient	Diet		SEM	P-value
	CON ^b	HI ^b		
Dry matter	0.818	0.832	0.321	0.002
Organic matter	0.867	0.869	0.003	0.658
Crude protein	0.792	0.773	0.006	0.002
Ether extract	0.973	0.971	0.002	0.538
Crude ash	0.279	0.282	0.010	0.914
Crude fiber	0.112	0.077	0.023	0.560
N-free extracts	0.907	0.929	0.031	< 0.001

^a n = 12 adult dogs per treatment ; six male and six female.

^b CON = control diet with lamb meal; HI = diet with *Hermetia illucens*.

3.3. D-glucosamine concentration in diet and feces and chitinase activity in feces

The analyzed concentration of D-glucosamine in the CON diet was 4.09 g/kg DM and in the HI diet 17.2 g/kg DM. The concentration of D-glucosamine was numerically higher ($P = 0.145$) in the feces of dogs fed the HI diet (92.2 ± 16 g/kg DM) compared to the CON diet (2.9 ± 0.4 g/kg DM), whereas the fecal chitinase activity was low and did not differ between both groups (0.12 ± 0.0 U; $P = 0.306$).

3.4. Microbial metabolites in fecal samples

The concentration of acetate ($P = 0.011$) and i-butyrate ($P = 0.049$) were higher and the concentration of propionate was numerical higher ($P = 0.053$) in the feces of dogs fed the CON diet compared to the HI diet (Table 5). No differences occurred in the concentration of L- and D-lactate in fecal samples of dogs fed either the CON or HI diet. The ammonium concentration was higher ($P = 0.001$) in the feces of dogs fed the CON diet (Table 5).

3.5. Complete and differential blood count

All data were in the reference range for adult dogs and no differences occurred between the HI and the CON diet (Table 6).

3.6. Peripheral blood lymphocyte phenotyping

The phenotyping of mononuclear cells in the peripheral blood did not show any differences between the two dietary periods (Table 7).

3.7. Lymphocyte proliferation test

The mitogen-induced stimulation of isolated mononuclear lymphocytes was not different between the CON and the HI diet (Table 8).

Also, the isolated mononuclear lymphocytes stimulated by different concentrations of feed antigen did not differ between the two feedings groups (Table 8).

4. Discussion

The present study shows that larvae meal from *Hermetia illucens* can be considered as an alternative protein source in dog nutrition. The data indicate no major differences of a HI containing diet (200 g/kg) in the CTTAD, immune function and fecal quality compared to a lamb meal based dog food. It should be noted that the control diet used as a reference is not only different in the protein source, but also in some other components and nutrient levels. This is a study limitation in the sense of a precise assessment of the insect protein, which however could not be avoided due to the production conditions. The decision to accept this is a restriction in comparability

Table 5

Concentration and proportion of short chain fatty acids and concentration of L-, D-lactate and ammonium in the feces of dogs fed the control diet and the diet containing *Hermetia illucens*^a.

	Diet		SEM	P-value
	CON ^c	HI ^c		
SCFA ^c , mmol/kg				
Acetate	97.5	77.1	3.46	0.011
Propionate	33.3	27.7	2.35	0.053
i-Butyrate	2.59	2.01	0.180	0.049
n-Butyrate	24.3	22.3	2.82	0.751
i-Valerate	4.08	3.57	0.270	0.222
n-Valerate	4.18	3.79	0.690	0.505
Total fatty acids	166	137	5.3	0.866
SCFA, mol. %				
Acetate	58.9	57.1	1.63	0.606
Propionate	19.8	20.4	1.28	0.706
i-Butyrate	1.55	1.48	0.100	0.400
n-Butyrate	14.6	15.7	1.74	0.729
i-Valerate	2.45	2.64	0.160	0.531
n-Valerate	2.69	2.67	0.460	0.906
L-Lactate, mmol/kg	1.45	1.01	0.223	0.312
D-Lactate, mmol/kg	1.01	0.89 ^b	0.165	0.605
Ammonium, mmol/kg	37.4	21.7	2.31	< 0.001

^a n = 12 adult dogs per treatment ; six male and six female.

^b n = 10 (two samples were below the detection limit).

^c CON = control diet with lamb meal; HI = diet with *Hermetia illucens*; SCFA = short chain fatty acids.

Table 6

Results of the complete blood count and the differential blood count of blood samples from dogs fed the control diet and the diet containing *Hermetia illucens*^a.

Item	Diet		SEM	P-value	Reference values ^c
	CON ^b	HI ^b			
Leukocytes (10 ⁹ /L)	8.29	8.75	0.314	0.386	6.0–12.0
Erythrocytes (10 ¹² /L)	7.09	6.99	0.092	0.571	5.5–8.5
Hemoglobin (g/L)	165	161	2.0	0.318	150–190
Hematocrit (L/L)	0.47	0.46	0.005	0.618	0.44–0.52
Thrombocytes (10 ⁹ /L)	265	273	16.2	0.633	150–500
Segmented neutrophils (10 ⁹ /L)	5.09	5.46	0.247	0.365	3.0–9.0
Lymphocytes (10 ⁹ /L)	2.28	2.38	0.078	0.203	1–3.6
Monocytes (10 ⁹ /L)	0.48	0.48	0.029	0.754	0.04–0.5
Eosinophils (10 ⁹ /L)	0.45	0.42	0.051	0.615	0.04–0.6

^a n = 12 adult dogs per treatment ; six male and six female.

^b CON = control diet with lamb meal; HI = diet with *Hermetia illucens*.

^c Laboklin GmbH & Co. KG, Bad Kissingen, Germany.

Table 7

Phenotyping of peripheral blood mononuclear cells of dogs fed the control diet and the diet containing *Hermetia illucens*^a.

Item	Diet		SEM	P-value
	CON ^c	HI ^c		
CD3 ⁺ , %	74.0	71.5	0.97	0.168
CD4 ⁺ , %	48.3 ^b	48.3	0.84	0.832
CD5 ⁺ , %	75.3	75.2	0.96	0.941
CD8β ⁺ , %	19.9	18.7	0.65	0.213
CD21 ⁺ , %	21.4	22.2	0.92	0.606
MHC II ⁺ , %	97.5 ¹	97.9 ^b	0.18	0.339

^a n = 12 adult dogs per treatment; six male and six female.

^b n = 11.

^c CON = control diet with lamb meal; HI = diet with *Hermetia illucens*.

Table 8

Mitogen induced and feed antigen induced proliferation (stimulation index) of isolated mononuclear lymphocytes of dogs fed the control diet and the diet containing *Hermetia illucens*^a.

Mitogen ^b	Diet		SEM	P-value
	CON ^b	HI ^b		
PWM, SI	9.81	11.4	0.76	0.365
PHA, SI	11.0	14.2	1.15	0.124
Concentration of HI protein, µg/mL				
30, SI	0.66	0.70	0.048	0.620
75, SI	0.91	0.99	0.067	0.358
125, SI	1.20	1.26	0.099	0.683
250, SI	2.06	2.19	0.241	0.526
500, SI	4.44	4.46	0.453	0.960

^a n = 12 adult dogs per treatment ; six male and six female.

^b CON = control diet with lamb meal; HI = diet with *Hermetia illucens*; PWM = pokeweed mitogen; PHA = phytohemagglutinin; SI = stimulation index.

between both diets. Lastly, it was also based on the fact that the reference diet has been on the market for many years and is used successfully in practice in many cases of feed intolerances in dogs. The test results also show that dogs do not appear to have a high capacity for the hydrolysis of chitin.

Both, the CON and the HI diets were well accepted and tolerated by all dogs. Dogs receiving the HI diet had a higher fecal score but both feeding groups had formed and dry feces and the dry matter of the feces did not differ between the feeding groups. The lower daily fecal output of dogs receiving the HI diet could be of benefit for dog owners, especially for owners of large breed dogs. The absence of clinical signs of dietary intolerance and the good palatability of the test diet indicates that the tested larvae meal can be considered as an alternative protein source in canine diets.

Compared to the CON diet with lamb meal, the HI diet had a higher CTTAD of dry matter and nitrogen-free extract, but a lower CTTAD of crude protein though the differences were small and not of practical relevance. Here it has to be kept in mind, that both experimental diets did not only differ regarding protein sources, but also in the other ingredients. The HI diet contained in descending

order rice meal, HI larvae meal, barley, and potato protein. Thus, protein was dominated by HI meal, but the other ingredients, especially the potato protein, have also contributed substantially. The total tract apparent digestibility of potato protein is not clear, as no specific data are available for dogs to our best knowledge. Assessing the total tract apparent digestibility data, obvious differences occurred also in the nutrient composition. The analyzed content of crude fiber in the HI diet was almost double than in the CON diet, while the concentration of TDF, insoluble and soluble fiber was comparable between the diets. However, it can be assumed that the fiber content in the HI diet was influenced by the high chitin content of the insect meal, as it is unclear to what extent chitin particles behave like a fibrous material and whether chitin also has negative effects on the digestibility of diets for dogs.

Ammonium concentrations in the feces of CON diet fed dogs were higher compared to the feeding of the HI diet. Fecal ammonium indicates the colonic fermentation of pre-cecal undigested protein. Pre-cecal apparent digestibility data could not be measured in this study, but a lower ileal and total tract apparent protein digestibility of lamb meal has been demonstrated in dogs, roosters and mink before, compared to meat and bone meal and poultry by-product meal (Johnson et al., 1998) or poultry meal and fish meal, respectively (Tjernsbekk et al., 2014). These findings seem to be related to the product quality, especially the ash content and the heat treatment, both of which are essential determinants of apparent protein digestibility. In studies with different protein carriers, there is the problem that not only the composition of the feed and the proportions of the respective components are important, but also the quality (muscle - collagen ratio) and especially the thermal treatment of the proteins. It is known that the sterilization of lamb meal in Europe must be carried out in accordance with the pertinent regulations, which means that sterilization is carried out in accordance with the Regulation (EC) No 1069/2009. Thus, sterilization procedures may result in a reduced pre-cecal protein digestibility. Due to the lack of cannulated dogs, this could not be further tested. This is a general problem that limits the comparability of diets and protein carriers due to possible differences in process technology.

Compared with an *in vitro* fermentation study with larvae meal from HI and dog feces (Bosch et al., 2014), the apparent organic matter digestibility was comparable while the *in vitro* trial showed a higher total tract apparent digestibility of crude protein/nitrogen compared to this *in vivo* study. However, the comparison of the apparent total tract digestibility of a complete dry food and *in vitro* digestibility of the pure larvae meal is of limited information.

The fecal pattern of microbial fermentation metabolites did not show distinct differences between the CON and HI diet. The higher concentration of acetate and the numerical higher concentration of total short-chain fatty acids in the feces of dogs fed the CON diet indicate a general higher fermentation in the large intestine and might be in line with the assumption, that the control diet had a lower pre-cecal digestibility. The distribution pattern of SCFA, including the branched chain fatty acids, was similar between the two feedings groups. The differences in the concentration of acetic acid and total short-chain fatty acids must also be interpreted against the background that the control diet contained sugar beet pulp as a fiber carrier which, as a fermentable fiber, can have an influence on microbial metabolites. In this respect, the differences are not necessarily due to the protein source but to the entire diet.

Amongst protein and fat, chitin is an important component of insects, with a high variability depending on species and development stage. Chitin is a polysaccharide composed of N-acetyl-D-glucosamine units. The polymer of N-acetyl-D-glucosamine residues linked by beta-(1,4)-glycosidic bonds is biologically relevant as it has been shown to be a potential target for the intestinal immune system (Elieh Ali Komi et al., 2018). The hydrolysis of chitin results in lower molecular weight oligosaccharides and chitosans, which would be a source of carbon and nitrogen (Cohen-Kupiec and Chet, 1998) and might even have a prebiotic activity in the intestinal tract (Borelli et al., 2017). The chitin concentration in the larvae meal from HI was estimated at 9.82 % in dry matter, based on the analysis of N-acetyl-D-glucosamine. This is in line with data available from literature (Kroeckel et al., 2012). In the HI diet, 17.2 g/kg D-glucosamine could be detected, which correlates well with the content of 200 g/kg larvae meal in the diet. In the CON diet, 4.1 g/kg D-glucosamine was determined in absence of added larvae meal. The origin is not clear but it can be supposed that the measured D-glucosamine derived from other polysaccharides which can be found for example in connective tissue such as cartilage (Thonar et al., 1985).

It is questionable if dogs or the intestinal microbiota can produce chitinase and if the chitinase activity will increase after an intake of chitin-containing food. Chitin degrading enzymes in the digestive tract can originate from the food, the microbiota or from the host organism (Simunek et al., 2001). The acidic mammalian chitinase (AMCase) was documented in the gastrointestinal system for mice, cattle, chicken, and humans (Boot et al., 2001; Suzuki et al., 2002). The occurrence of chitinase activity in the canine intestinal tract has not yet been described. The fecal chitinase activity was equally low with both dietary treatments. It can be assumed that the higher chitin content in the HI diet did not stimulate the chitinase activity in the intestine. Apparently, the higher content of chitin in the HI diet did not have a clear negative influence on the apparent total tract nutrient digestibility. Whether chitin was utilized by bacteria or the host is not clear from the data.

It is not clear if and how insects may influence the hosts' immune system, however, this would be an important aspect for their future use. For the diagnosis of suspected adverse food reactions several tests are available. The sensitivity of most of these tests, especially the analysis of serum antibodies against food antigens seems to be very low (Mueller and Tsohalis, 1998; Bethlehem et al., 2012) and the clinical standard procedure still is the feeding of an elimination diet, eventually followed by food challenge (Verlinden et al., 2006). Food antigen induced lymphocyte proliferation test might be an alternative as it showed a high correlation (82 %) with oral food challenge (Ishida et al., 2004), but to our best knowledge, it has not been tested in dogs fed diets with insect protein.

A general problem in nutritional studies in which immunological reactions to food proteins are to be measured is the duration of the experiments. In this case, the time between blood samplings was five weeks and a cross-over design was chosen. There was no evidence that carry-over effects occurred. Nevertheless, it is mandatory to carry out corresponding investigations in immunological naive dogs. The dogs used in this experiment have not been in contact with the protein sources investigated in any way before the experiment, so it can be assumed that they were actually "naive".

The phenotyping of peripheral blood lymphocytes resulted in comparable distributions between the two dietary groups. This was

expected as there are no reported feed depending differences in the B- and T-lymphocyte population in former studies (Zinn et al., 2009). In addition, the complete and differential blood counts showed no differences between the dietary periods.

Neither the mitogen nor the feed antigen induced lymphocyte proliferation test was influenced by the two diets. The test is assessed to be positive to a feed antigen if SI is above 2 (Masuda et al., 2002; Ishida et al., 2004). Up to a larvae protein concentration of 125 µg/mL the SI was below 2 in the CON and HI diet and increased with concentrations of 250 and 500 µg/mL of extracted protein from HI larvae meal. The increase was equal in both feeding groups, even for the dogs receiving the diet with lamb meal, although they were not exposed to the HI diet before. This could be due to cross-reactions with other antigens or non-specific stimulation or due to an autofluorescence of non-vital cells. During the analysis by flow cytometry a shift of the whole cell population above the y-axis was recognized for the two highest protein concentrations and it wasn't possible to distinguish between BrdU positive (proliferated) and BrdU negative (non-proliferated) cells anymore. In addition, non-vital cells can bind the anti-BrdU antibody, they will be detected by flow cytometry and might therefore lead to false-positive results. To confirm this assumption, the proliferation tests were repeated for both diets without the addition of BrdU which makes a binding of the anti-BrdU antibody to BrdU impossible. As a fluorescence was still measured, it can be assumed that the SI > 2 with the high concentrations of the protein extract (250 and 500 µg/mL) are false-positive results due to autofluorescent non-vital cells. This can be supported by the fact, that none of the animals showed signs of food sensitivity during the feeding trial which would have appeared if a food allergy was present.

5. Conclusions

A diet containing 200 g/kg *Hermetia illucens* larvae meal did not result in major effects on the apparent total tract nutrient digestibility, fecal parameters or immunological data compared to a commercial dry diet with lamb meal. The comparability of the two diets used is limited by the use of different components, so that the results must be interpreted with caution. Despite these limitations, it can be assumed, that larvae meal from *Hermetia illucens* is an alternative protein source in dog nutrition.

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRedit authorship contribution statement

Susan Kröger: Writing - original draft, Investigation. **Carolyn Heide:** Investigation, Formal analysis. **Jürgen Zentek:** Supervision, Writing - review & editing, Conceptualization, Formal analysis.

Declaration of Competing Interest

None.

Acknowledgement

We would like to thank the animal care takers and the technical assistance of the Institute of Animal Nutrition. In particular, we would like to thank Anett Kristen for her assistance and support for the analysis of chitin and chitinase enzyme activity.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.anifeeds.2020.114699>.

References

- Barroso, F.G., de Haro, C., Sanchez-Muros, M.J., Venegas, E., Martinez-Sanchez, A., Perez-Banon, C., 2014. The potential of various insect species for use as food for fish. *Aquaculture* 422, 193–201. <https://doi.org/10.1016/j.aquaculture.2013.12.024>.
- Bethlehem, S., Bexley, J., Mueller, R.S., 2012. Patch testing and allergen-specific serum IgE and IgG antibodies in the diagnosis of canine adverse food reactions. *Vet. Immunol. Immunopathol.* 145, 582–589. <https://doi.org/10.1016/j.vetimm.2012.01.003>.
- Boot, R.G., Blommaert, E.F.C., Swart, E., Ghauharali-van der Vlugt, K., Bijl, N., Moe, C., Place, A., Aerts, J.M.F.G., 2001. Identification of a novel acidic mammalian chitinase distinct from chitotriosidase. *J. Biol. Chem.* 276, 6770–6778. <https://doi.org/10.1074/jbc.M009886200>.
- Borelli, L., Coretti, L., Dipineto, L., Bovera, F., Menna, F., Chiariotti, L., Nizza, A., Lembo, F., Fioretti, A., 2017. Insect-based diet, a promising nutritional source, modulates gut microbiota composition and SCFAs production in laying hens. *Sci. Rep.* 7, 6269. <https://doi.org/10.1038/s41598-017-16560-6>.
- Bosch, G., Zhang, S., Oonincx, D.G.A.B., Hendriks, W.H., 2014. Protein quality of insects as potential ingredients for dog and cat foods. *J. Nutr. Sci.* 3 (e29), 1–4. <https://doi.org/10.1017/jns.2014.23>.
- Bradford, M.M., 1976. A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. *Anal. Biochem.* 7, 248–254.
- Cohen-Kupiec, R., Chet, I., 1998. The molecular biology of chitin digestion. *Curr. Opin. Biotechnol.* 9, 270–277. [https://doi.org/10.1016/S0958-1669\(98\)80058-X](https://doi.org/10.1016/S0958-1669(98)80058-X).
- Elieh Ali Komi, D., Sharma, L., Dela Cruz, C.S., 2018. Chitin and its effects on inflammatory and immune responses. *Clin. Rev. Allergy Immunol.* 54, 213–223. <https://doi.org/10.1007/s12016-017-8600-0>.
- Ishida, R., Masuda, K., Kurata, K., Ohno, K., Tsujimoto, H., 2004. Lymphocyte blastogenic responses to inciting food allergens in dogs with food hypersensitivity. *J. Vet. Internal Med.* 18, 25–30. [https://doi.org/10.1016/S0165-2427\(01\)00337-3](https://doi.org/10.1016/S0165-2427(01)00337-3).

- Johnson, M.L., Parson, C.M., Fahey Jr., G.C., Merchen, N.R., Aldrich, C.G., 1998. Effects of species raw material source, ash, content, and processing temperature on amino acid digestibility of animal by-product meals by cecectomized roosters and ileally cannulated dogs. *J. Anim. Sci.* 76, 1112–1122. <https://doi.org/10.2527/1998.7641112x>.
- Kroeckel, S., Harjes, A.G.E., Roth, I., Katz, H., Wuertz, S., Susenbeth, A., Schulz, C., 2012. When a turbot catches a fly: evaluation of a pre-pupae meal of the Black Soldier fly (*Hermetia illucens*) as fish meal substitute – growth performance and chitin degradation in juvenile turbot (*Psetta maxima*). *Aquaculture* 364–365, 345–352. <https://doi.org/10.1016/j.aquaculture.2012.08.041>.
- Kröger, S., Vahjen, W., Zentek, J., 2017. Influence of lignocellulose and low or high levels of sugar beet pulp on nutrient digestibility and the fecal microbiota in dogs. *J. Anim. Sci.* 95, 1598–1605. <https://doi.org/10.2527/jas2016.0873>.
- Masuda, K., Sakaguchi, M., Saito, S., Deboer, D.J., Yamashita, K., Hasegawa, A., Ohno, K., Tsujimoto, H., 2002. Seasonal atopic dermatitis in dogs sensitive to a major allergen of Japanese cedar (*Cryptomeria japonica*) pollen. *Vet. Dermatol.* 13, 55–61.
- Middelbos, I.S., Fastinger, N.D., Fahey Jr, G.C., 2007. Evaluation of fermentable oligosaccharides in diets fed to dogs in comparison to fiber standards. *J. Anim. Sci.* 85, 3033–3044. <https://doi.org/10.2527/jas.2007-0080>.
- Mueller, R., Tsohalis, J., 1998. Evaluation of serum allergen-specific IgE for the diagnosis of food adverse reactions in the dog. *Vet. Dermatol.* 9, 167–171. <https://doi.org/10.1046/j.1365-3164.1998.00107.x>.
- National Research Council (NRC), 2006. *Nutrient Requirements of Dogs and Cats*. National Academic Press, Washington DC, USA.
- Naumann, K., Bassler, R., 2004. *Methodenbuch Band III: Die Chemische Untersuchung Von Futtermitteln*. Melsungen, Neumann-Neudamm, Germany.
- Newton, G.L., Booram, C.V., Barker, R.W., Hale, O.M., 1977. Dried *Hermetia Illucens* larvae meal as a supplement for swine. *J. Anim. Sci.* 44, 395–400. <https://doi.org/10.2527/jas1977.443395x>.
- Oonincx, D.G.A.B., van Isterbeeck, J., Heetkamp, M.J.W., van den Brand, H., van Loon, J.J.A., van Huis, A., 2010. An exploration on greenhouse gas and ammonia production by insect species suitable for animal or human consumption. *PLoS One* 5, e14445. <https://doi.org/10.1371/journal.pone.0014445>.
- Rumpold, B.A., Schlüter, O.K., 2013. Nutritional composition and safety aspects of edible insects. *Mol. Nutr. Food Res.* 57, 802–823. <https://doi.org/10.1002/mnfr.201200735>.
- Salomone, R., Saija, G., Mondello, G., Giannetto, A., Fasulo, S., Savastano, D., 2017. Environmental impact of food waste bioconversion by insects: application of life cycle assessment to process using *Hermetia illucens*. *J. Clean. Prod.* 140, 890–905. <https://doi.org/10.1016/j.jclepro.2016.06.154>.
- Shen, C.R., Chen, Y.S., Yang, C.J., Chen, J.K., Liu, C.L., 2010. Colloid chitin azure is a dispersible, low-cost substrate for chitinase measurements in a sensitive, fast, reproducible assay. *J. Biomol. Screen.* 15, 213–217. <https://doi.org/10.1177/1087057109355057>.
- Short, F.J., Gorton, P., Wiseman, J., Boorman, K.N., 1996. Determination of titanium dioxide added as an inert marker in chicken digestibility studies. *Anim. Feed Sci. Technol.* 59, 215–221.
- Simunek, J., Hodrova, B., Bartonova, H., Kopečný, J., 2001. Chitinolytic bacteria of the mammal digestive tract. *Folia Microbiol. (Praha)* 46, 76–78.
- SPSS, 2017. *SPSS Statistics Version 25.0*. IBM, Chicago, IL, USA.
- Suzuki, M., Fujimoto, W., Goto, M., Morimatsu, M., Syuto, B., Iwanaga, T., 2002. Cellular expression of gut chitinase mRNA in the gastrointestinal tract of mice and chickens. *J. Histochem. Cytochem.* 50, 1081–1089. <https://doi.org/10.1177/002215540205000810>.
- Thonar, E.J., Lenz, M.E., Klintworth, G.K., Caterson, B., Pachman, L.M., Glickman, P., Katze, R., Huff, J., Kuettner, K.E., 1985. Quantification of keratan sulfate in blood as a marker of cartilage catabolism. *Arthritis Rheum.* 28, 1367–1376. <https://doi.org/10.1002/art.1780281209>.
- Tjersbekk, M.T., Tauson, A.H., Ahlstrom, O., 2014. Ileal, colonic and total tract nutrient digestibility in dogs (*Canis familiaris*) compared with total tract digestibility in mink (*Neovison vison*). *Arch. Anim. Nutr.* 68, 245–261. <https://doi.org/10.1080/1745039X.2014.915137>.
- Verlinden, A., Hesta, M., Millet, S., Janssens, G.P.J., 2006. Food allergy in dogs and cats: a review. *Crit. Rev. Food Sci. Nutr.* 46, 259–273. <https://doi.org/10.1080/10408390591001117>.
- Zinn, K.E., Hernot, D.C., Fastinger, N.D., Karr-Lilienthal, L.K., Bechtel, P.J., Swanson, K.S., Fahey, G.C., 2009. Fish protein substrates can substitute effectively for poultry by-product meal when incorporated in high-quality senior dog diets. *J. Anim. Physiol. Anim. Nutr. (Berl)* 93, 447–455. <https://doi.org/10.1111/j.1439-0396.2008.00826.x>.